AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

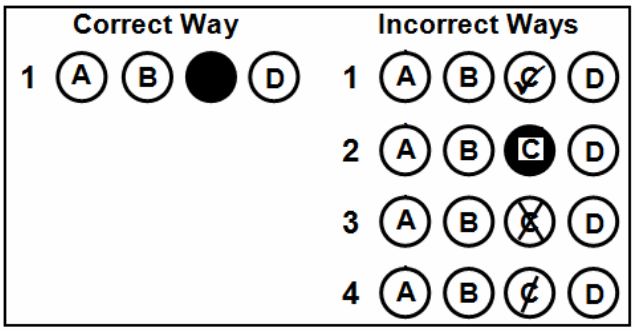
MAY 2014

Biology Paper I

Time: 35 minutes Marks: 25

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 25 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.

1. Respiration is different from gaseous exchange as it

   A. involves lungs and its associated structures.
   B. takes place in higher multicellular organisms.
   C. requires a concentration gradient to occur.
   D. consists of a series of chemical reactions.

2. A person is showing the following symptoms of a respiratory disorder.

   i. High fever
   ii. Cough with sputum
   iii. Lungs filled with pus

   Which of the following disorders is the person suffering from?

   A. Emphysema
   B. Pneumonia
   C. Lung cancer
   D. Asthma

3. Which of the following options CORRECTLY matches the kind of plant with its waste product?

	Kind of Plant	**Waste Product**
A.	Rubber plant	Calcium oxalate
B.	Tomato	Gum
C.	Lady finger	Mucilage
D.	*Keekar*	Latex

4. Kidneys keep the composition of blood constant in all of the following ways EXCEPT through

   A. the removal of nitrogenous waste from the body.
   B. ultrafiltration of blood in glomerulus.
   C. the release of adrenaline through adrenal glands.
   D. selective reabsorption of water by loop of Henle.

5. In a healthy individual, the substance that is excreted in urine in the smallest amount is

   A. glucose.
   B. sodium ions.
   C. urea.
   D. water.

6. The function of medulla oblongata is to

   A. regulate body temperature.
   B. secrete different hormones.
   C. bring about movement of eye balls.
   D. control involuntary actions.

7. The function of myelin sheath in a neuron is to

A. serve as an insulating material.
B. provide nourishment for the fibre.
C. speed up the transmission of impulses.
D. generate adenosine triphosphate molecules.

8. The diagram shows two different states (R and S) of the pupil in a human eye.



R S

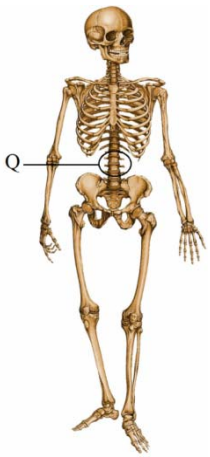
The state **R** of the pupil shows that a person

A. is looking at a distant object.
B. is looking at a nearby object.
C. has entered a dark room.
D. has entered a bright room.

9. Which of the following is present in almost all types of connective tissues, specially bones and cartilage?

A. Vitamin D
B. Red blood cells
C. Collagen fibres
D. Calcium sulphate

10. The diagram shows a human skeleton.



The part of the skeleton encircled as 'Q' bears

A. ball and socket joint.
B. hinge joint.
C. slightly moveable joint.
D. immoveable joint.

11. What is the function of tendons in the body?

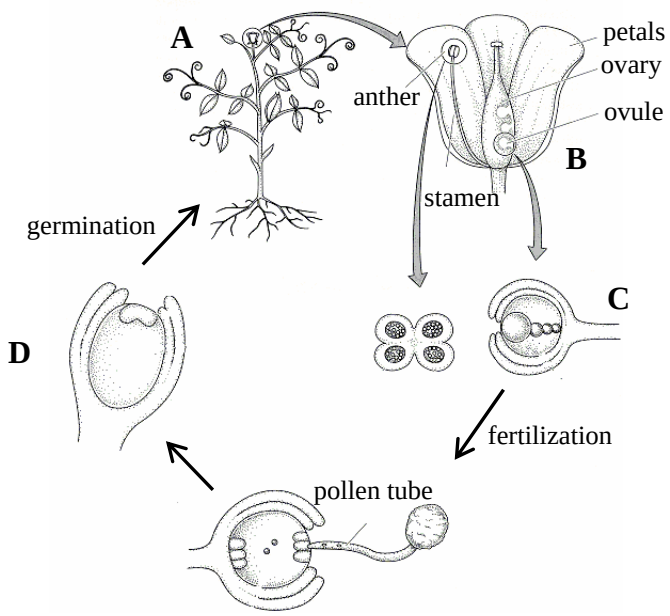
A. Join one bone to another
B. Reduce friction between the bones
C. Attach muscles to the bones
D. Conduct nerve impulses to the bones

12. The plant in the given diagram shows natural vegetative propagation through



A. roots.
B. stem.
C. leaves.
D. flower.

13. The diagram shows some stages of the life cycle of a flowering plant. At which of the labelled stages of the life cycle could a triploid nucleus (3n) be present?



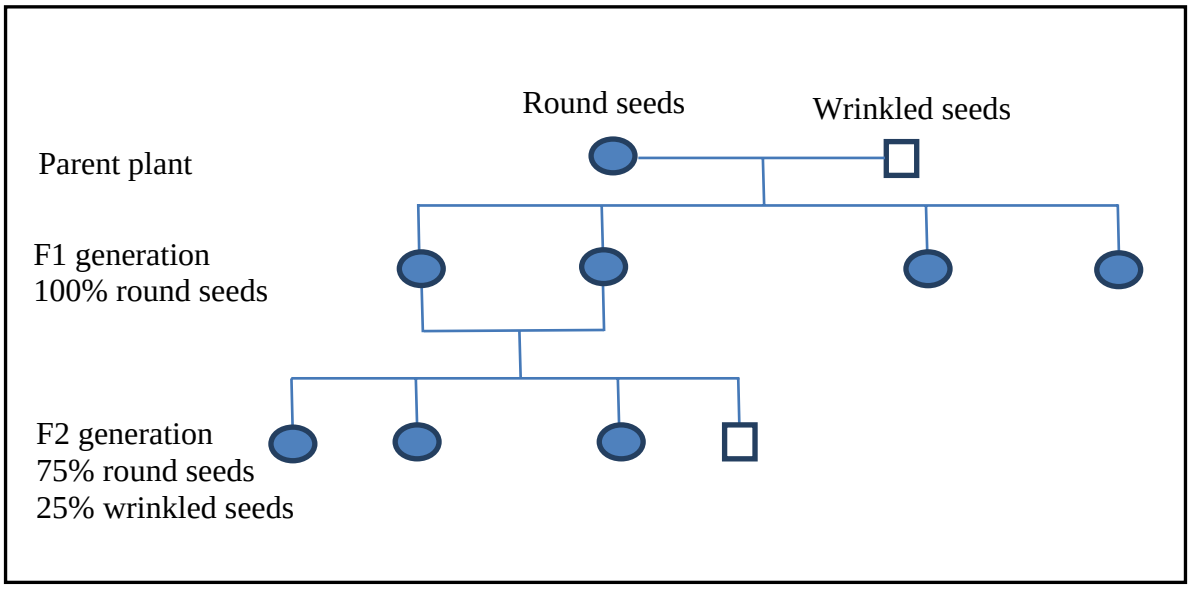
14. Which of the following statements is CORRECT about internal fertilization?

A. It involves release of gametes into the surroundings.
B. It causes protection of the developing embryo.
C. It takes place mostly in aquatic environments.
D. It requires large number of gametes to be released.

15. A heterozygous black fur cat is crossed with a heterozygous black fur cat. Which of the following percentage genotypes may NOT appear in the F1 generation?

A. 50% Bb
B. 25% BB
C. 50% bb
D. 25% bb

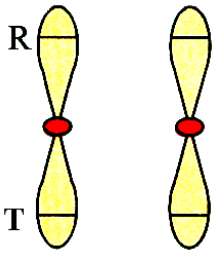
16. The pedigree shows crosses between different seeds with their expected outcomes.



Wrinkled seeds do not appear in F1generation but they appear in F2 generation because

A. round seeds produce offspring without crossing with wrinkled seeds.
B. the genotype of round seeds in parent plant is RR.
C. recessive characters always appear alternately.
D. wrinkled seeds are dominant over round seeds.

17. The diagram shows a pair of homologous chromosomes with genes.



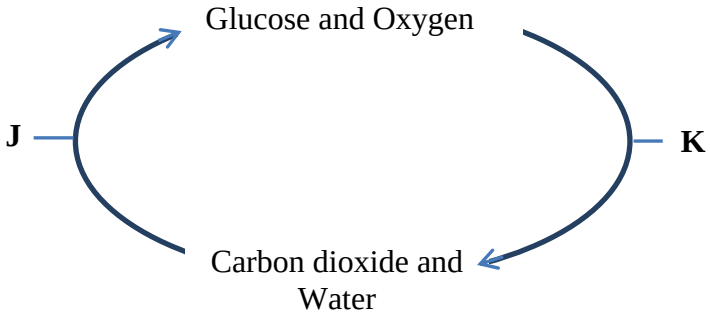
Which of the following is the allele for gene **T**?

A. U
B. t
C. r
D. R

18. A species is a group of organisms which

A. consists of unicellular organisms only.
B. lives without abiotic factors.
C. interbreed freely in nature.
D. develop parasitism among themselves.

19. Look at the given diagram carefully.



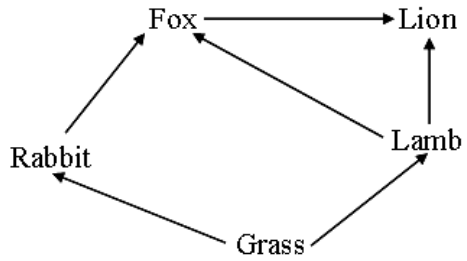
Which of the following CORRECTLY identifies forms of energy at point J and K in the given cycle?

	Point J	**Point K**
A.	Heat energy	Heat energy
B.	Light energy	Light energy
C.	Heat energy	Light energy
D.	Light energy	Heat energy

20. Crabs usually attach themselves to the surface of large fish. In this way large fish provide easy transport to crabs to better feeding places. This is an example of

A. parasitism.
B. mutualism.
C. commensalism.
D. predation.

21. The diagram shows a food web of a particular geographical location.



A team of hunters killed most of the rabbits there.
What will be the effect of killing rabbits on the food web?

A. There will be more grass.
B. The amount of grass will decrease.
C. The number of foxes will increase.
D. There will be more lions.

22. All of the following are objectives of genetic engineering EXCEPT

A. isolating particular genes.
B. discouraging the use of vaccines.
C. treating genetic defects.
D. producing improved drugs.

23. In an industrial fermenter, an impeller is a device which

A. removes heat during microbial activities.
B. mixes air bubbles with nutrient broth.
C. controls pH of the culture broth.
D. monitors temperature inside the fermenter.

24. The two important groups of antibiotics are

A. sulphonamides and tetracyclines.
B. sulphonamides and hallucinogens.
C. tetracyclines and sedatives.
D. sedatives and hallucinogens.

25. Which of the following is CORRECT about marijuana?

	Causes Hallucination	**Stimulant**	**Addiction**	**Decreases Production of Sperms in Men**
A.	✓	✓	✓	✗
B.	✓	✗	✓	✓
C.	✓	✗	✗	✓
D.	✗	✓	✓	✗

END OF PAPER

Please use this page for rough work